Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3391-WE01

Title: (BEng/MEng)
Control and Signal Processing 3

Time Allowed:	2 hours
Additional Material provided:	Graph Paper
Materials permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	
	This is an in-person, invigilated exam. Answer ALL questions. Write your answer in the white-covered answer booklet with barcodes. All relevant workings must be shown.

Revision:	

Section A

Q.1

(a) Consider the interconnection of linear time invariant (LTI) systems as shown in Figure Q.1.a. Express the overall impulse response h[n] in terms of the individual impulse responses $h_1[n]$, $h_2[n]$, $h_3[n]$, and $h_4[n]$.

[10%]

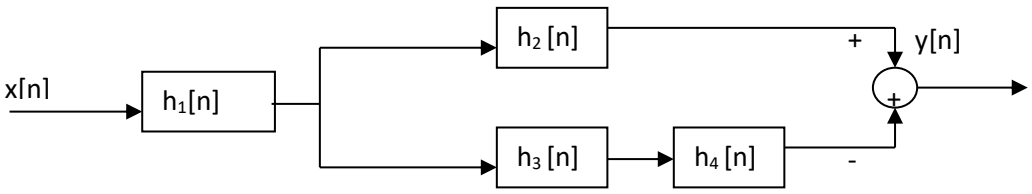


Figure Q.1.a

Determine h[n] when

$h_1[n] = \{0.5, 0.25, 0.5\}$ $\quad 0 \leq n \leq 2$

$h_2[n] = h_3[n] = (n+1)u[n]$

$h_4[n] = \delta[n-2]$

where u[n] is the unit step function.

[30%]

(b) Discuss two practical methods for determining the impulse response of a linear time invariant system.

[15%]

(c) Discuss the conditions for distortion-less transmission

[10%]

(d) For the cascade system in Figure Q.1.d in next page

(i) Find the spectrum of z(t) in terms of the spectrum of x(t),

[30%]

(ii) Find the spectrum of z(t) for the special case of x(t) with no DC component.

[5%]

continued

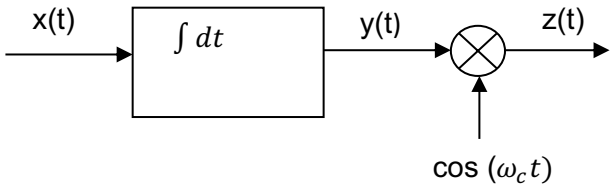


Figure Q.1.d

Q.2

a) Discuss the following operations and illustrate graphically:

(i) Time compression of a signal and the effect on its spectrum.

[10%]

(ii) Time reversal and time delay of a signal

[10%]

b) Find the impulse response of the linear causal system with input x[n] and corresponding output y[n] given by

$$y[n] - y[n-1] = x[n] - x[n-3].$$

[10%]

c) The system y[n] is given by

y[n] = a. x[3n−2]

Characterise the properties of this system in terms of linearity, causality and time-invariance.

[15%]

d) The output y(t) of a causal linear time invariant system is related to its input x(t) by the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

(i) Determine the frequency response of the system

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

[10%]

continued

(ii) If $x(t) = e^{-t}u(t)$, determine $Y(\omega)$, the Fourier transform of the output

[10%]

(iii) Determine $Y(\omega)$, if the input

$$X(\omega) = \frac{1}{(2 + j\omega)(1 + j\omega)}$$

[10%]

e) A vehicle is travelling at 40 miles/hour in a 30 miles/hour zone. Continuous waveform (CW) radar operating at 900 MHz is used to detect vehicles exceeding the speed limit. Assuming that the radar makes an angle α with the road, and detects the change in frequency of the received signal f_d given by

$$f_d = \pm \frac{2v}{\lambda} cos\alpha$$

where λ is the wavelength of the CW signal.

(i) What is the optimum value of the angle α to determine the speed of the vehicle?

[5%]

(ii) When converting the received signal to digital, what is the minimum sampling frequency to prevent aliasing?

[15%]

(iii) Assuming that the digital processor uses an algorithm that requires a sampling rate of power of 2. Determine the minimum required sampling rate.

[5%]

continued

Section B

Q.3

A system plant is described by the following transfer function:

$$G(s) = \frac{5}{s^2 + 4s + 3}$$

and is placed in a control loop as shown in Figure Q.3.

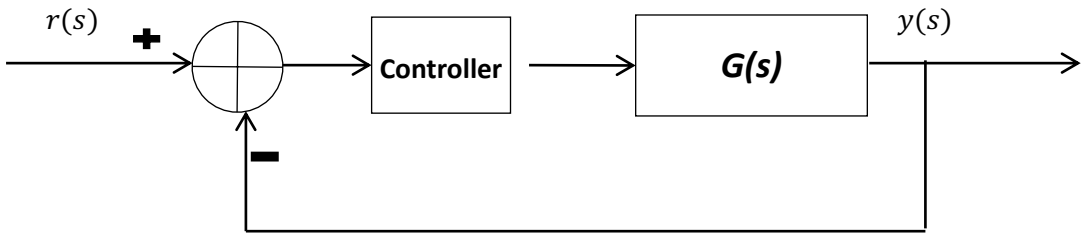


Figure Q.3: A closed-loop control system with negative unit feedback.

A controller $D(s)$ is required to satisfy a performance specification of damping ratio $\zeta = 0.6$ and undamped natural frequency $\omega_n = 5$ rad/sec.

(a) On the graph paper provided, draw the root locus of $G(s)$ and give the procedures and steps of the root locus plotting.

[20%]

(b) Mark the position of the required closed loop poles to meet the performance specification of damping ratio $\zeta = 0.6$ and undamped natural frequency $\omega_n = 5$ rad/sec. State whether it is feasible to use a gain compensator to meet the performance specification and state the reasons.

[20%]

(c) If the controller in Figure Q.3 is a practical phase-lead compensator $D(s) = k\frac{s+z}{s+p}$, design this phase-lead compensator to satisfy the performance specification of damping ratio $\zeta = 0.6$ and undamped natural frequency $\omega_n = 5$ rad/sec.

[40%]

(d) State the different effects when discretising a controller with the Backward Difference and Forward Difference methods.

[20%]

continued

Q.4

(a) State the effects of changing each term of proportional-integral-derivative (PID) controller (i.e., P term, I term, and D term) on the system performance, and compare PID/PI/PD controllers with phase-lead/lag and lead-lag compensators.

[20%]

(b) A closed-loop control system is shown in Figure Q.4 where the plant's transfer function is given by $G(s) = \frac{12}{s(s+2)(s+3)}$.

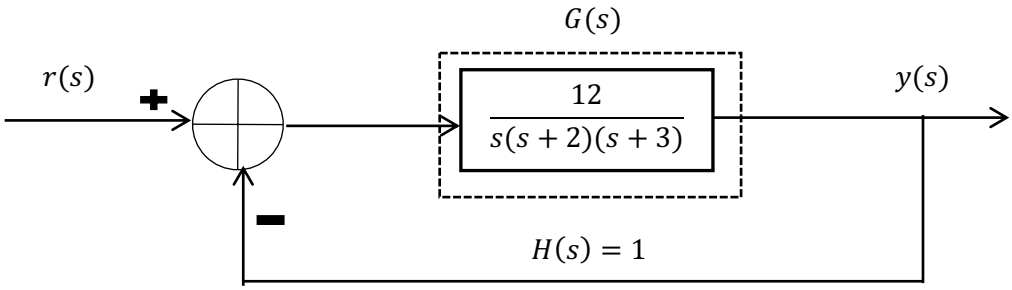


Figure Q.4: A control system with negative unit feedback

(i) Write the open-loop frequency response formula $GH(i\omega)$ in the complex number form of $a + ib$.

[20%]

(ii) Calculate its open-loop frequency response as ω varies, and present your data using a Nyquist diagram.

[20%]

(iii) Find the gain margin, the phase margin, the phase cross-over frequency, and the gain cross-over frequency.

[20%]

(iv) Discuss potential methods for improving the system stability and use numerical examples to demonstrate their impacts on the system stability.

[20%]

continued

L3 Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) = a_o + \sum_{k=1}^{\infty} \left[a_k \cos k\omega_o t + b_k \sin k\omega_o t \right]$$

where

$$a_o = \frac{1}{T} \int_T x(t)dt = dc\ term$$

$$a_k = \frac{2}{T} \int_T x(t) \cos k\omega_o t dt \qquad b_k = \frac{2}{T} \int_T x(t) \sin k\omega_o t dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) = a_o + \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2} \cos(k\omega_o t + \phi_k)$$

where

$$\phi_k = -\tan^{-1} \frac{b_k}{a_k}$$

Complex exponential form

$$x(t) = \sum_{k=-\infty}^{\infty} d_k\, e^{jk\omega_o t}$$

where $d_k = \frac{1}{T} \int_T x(t).e^{-jk\omega_o t} dt$

and $d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

continued

Fourier transform

$$\mathcal{F}\{x(t)\} = X(\omega)$$
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Alternatively it can be written in terms of the frequency in cycles per seconds or Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

or

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

Fourier transform properties

1. Linearity

$$x_1(\mathrm{t}) \leftrightarrow X_1(\omega), \text{and } x_2(\mathrm{t}) \leftrightarrow X_2(\omega) \text{ then}$$
$$\mathrm{a}x_1(\mathrm{t}) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$$

2. Differentiation

$$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. Integration

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

4. Time scaling

$$x(b\,\mathrm{t}) \leftrightarrow \frac{1}{|b|}X\left(\frac{\omega}{b}\right)$$

5. Time reversal

continued

$$x(-\mathrm{t}) \leftrightarrow X(-\omega) = X^{*}(\omega)$$

6. Time delay

$$y(t) = x(\mathrm{t}-\mathrm{T}) \leftrightarrow X(\omega)\, \mathrm{e}^{-\mathrm{j}\omega \mathrm{T}}$$

7. Multiplication in time

$$x_1(\mathrm{t}).x_2(\mathrm{t}) \leftrightarrow \frac{X_1(\omega) * X_2(\omega)}{2\pi}$$

8. Multiplication in frequency (convolution in time)

$$X_1(\omega)X_2(\omega) \leftrightarrow x_1(\mathrm{t}) * x_2(\mathrm{t})$$

9. Modulation property

$$x(\mathrm{t})\cos(\omega_o t) \leftrightarrow \frac{1}{2}X(\omega - \omega_{\mathrm{o}}) + \frac{1}{2}X(\omega + \omega_o)$$

10. Duality property

If $x(t) \leftrightarrow X(\omega)$

Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series

DFS: $d_k = X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{\frac{-j2\pi kn}{N}}$

Inverse Discrete Fourier Series

IDFS: $x[n] = \sum_{k=0}^{N-1} d_k\, e^{\frac{j2\pi kn}{N}}$

Euler's identities

$$A\sin(\omega_o t) = \frac{A}{j2}\left(e^{j\omega_o t} - e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t) = \frac{A}{2}\left(e^{j\omega_o t} + e^{-j\omega_o t}\right)$$

continued

$$A\cos(\omega_o t + \phi_o) = \frac{A}{2}\left(e^{j\omega_o t}e^{j\phi_o} + e^{-j\omega_o t}e^{-j\phi_o}\right)$$

$e^{\pm jx} = \cos x \pm j \sin x$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k]x_2[k]$$

Convolution with an impulse

$x(t) * \delta(t - t_o) = x(t - t_o)$

Fourier transform of an impulse

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$\delta(\omega) \leftrightarrow 1/2\pi$

Roots of a second order polynomial

continued

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

cos θ = cos(-θ) = sin(π/2 - θ)

sin θ = - sin(-θ) = cos(π/2 - θ)

tan θ = - tan(-θ) = cot(π/2 - θ)

sin 2θ = 2 cos θ sin θ

cos 2θ = $\cos^2\theta$ - $\sin^2\theta$ = 2 $\cos^2$ θ- 1 = 1 - 2 $\sin^2\theta$

tan 2θ = 2 tan θ/ (1 - tan2θ)

$$\sin\alpha.\sin\beta = \frac{1}{2}\{(\cos(\alpha - \beta) - cos(\alpha + \beta)\}$$

$$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha - \beta) + cos(\alpha + \beta)\}$$

$$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha - \beta) + sin(\alpha + \beta)\}$$

sin θ/2 = ± √[(1 - cos θ)/2]

cos θ/2 = ± √[(1 + cos θ)/2]

tan θ/2 = sin θ/(1 + cos θ)

$\cos^2$ θ+ $\sin^2$ θ = 1

$\sec^2$ θ - $\tan^2$ θ = 1

cosec^2 θ - $\cot^2$ θ = 1

$\sin^2$ θ = ½(1 - cos 2θ)

$\cos^2$ θ = ½(1 + cos 2θ)

$\tan^2$ θ = (1 - cos 2θ) /(1 + cos 2θ)

sin(A + B) = sin A cos B + cos A sin B

sin(A - B) = sin A cos B - cos A sin B

continued

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\tan(A + B) = (\tan A + \tan B)/(1 - \tan A \tan B)$

$\tan(A - B) = (\tan A - \tan B)/(1 + \tan A \tan B)$

$\sin A + \sin B = 2 \sin ½(A + B) \cos ½(A - B)$

$\sin A - \sin B = 2 \cos ½(A + B) \sin ½(A - B)$

$\cos A + \cos B = 2 \cos ½(A + B) \cos ½(A - B)$

$\cos A - \cos B = - 2 \sin ½(A + B) \sin ½(A - B)$

$\tan A + \tan B = \sin (A + B)/(\cos A \cos B)$

$\tan A - \tan B = \sin (A - B)/(\cos A \cos B)$

$\sin2A + \sin2B = 1 - \cos(A + B)\cos(A - B)$

$\sin2A - \sin2B = \sin(A + B)\sin(A - B)$

$\cos2A + \sin2B = 1 - \sin(A + B)\sin(A - B)$

$\cos2A - \sin2B = \cos(A + B)\cos(A - B)$

$\cos2A + \cos2B = 1 + \cos(A + B)\cos(A - B)$

$\cos2A - \cos2B = - \sin(A + B)\sin(A - B)$

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

$a2 = b2 + c2 - 2\ b\ c \cos A$

$a/\sin A = b/\sin B = c/\sin C.$

$(a - b)/(a + b) = \tan ½(A - B)/ \tan ½(A + B)$

continued

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

7 Steps for Root Locus Sketching:

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

continued

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle *Φ*

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\phi = \frac{(2q+1)}{n-m}\pi,$$

where $q = 0,1,\ldots,(n-m-1)$

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n}|s - p_i|}{\prod_{j=1}^{m}|s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n}\phi_{p_i} - \sum_{j=1}^{n}\phi_{z_i} = (2q+1)\pi$$

END